INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Exam 1 Review Worksheet Solutions

Counting

1. Let $U = \{1, 2, 3, \ldots, 12\}$ be the universal set. Let A be the set of multiples of 3 in U, and let B be the set of odd numbers in U.

   (a) Write A and B by listing their elements.

   Solution: $A = \{3, 6, 9, 12\}$ and $B = \{1, 3, 5, 7, 9, 11\}$.

   (b) Find $A \cup B$, $A \cap B$, and A^{c}.

   Solution:

   $$A \cup B = \{1, 3, 5, 6, 7, 9, 11, 12\}, \quad A \cap B = \{3, 9\}, \quad A^{\mathrm{c}} = \{1, 2, 4, 5, 7, 8, 10, 11\}.$$

   (c) Find $(A \cup B)^{\mathrm{c}}$ and $|A \cup B|$.

   Solution: $(A \cup B)^{\mathrm{c}} = \{2, 4, 8, 10\}$, the elements of U that are in neither A nor B, and $|A \cup B| = 8$.

   (d) Are A and B disjoint? Explain.

   Solution: No, A and B are not disjoint, since $A \cap B = \{3, 9\} \neq \emptyset$.

2. A sandwich shop lets you choose one of 4 breads, one of 5 fillings, and one of 3 cheeses, or no cheese at all. How many different sandwiches are possible?

Solution: There are 4 choices for the bread and 5 choices for the filling. Counting "no cheese" as one more option, there are $3+1 = 4$ choices for the cheese. By the Fundamental Principle of Counting, there are

$$4 \cdot 5 \cdot 4 = 80$$

possible sandwiches.

3. Compute each of the following without a calculator.

(a) $\dfrac{9!}{7!\,2!}$

Solution: $\dfrac{9!}{7!\,2!} = \dfrac{9 \cdot 8 \cdot 7!}{7!\,2!} = \dfrac{9 \cdot 8}{2 \cdot 1} = 36.$

(b) $0! + 1!$

Solution: $0! + 1! = 1 + 1 = 2.$

(c) $(2+3)!$ and $2! + 3!$. Are they equal?

Solution: $(2+3)! = 5! = 120$, whereas $2! + 3! = 2 + 6 = 8$. They are not equal.

4. Eight runners compete in a race. Gold, silver, and bronze medals go to the first, second, and third place finishers. In how many ways can the medals be awarded, assuming there are no ties?

Solution: The order of the first three finishers matters, so each way of awarding the medals is a 3-permutation of the 8 runners. Therefore, there are

$$_8P_3 = \frac{8!}{(8-3)!} = \frac{8!}{5!} = 8 \cdot 7 \cdot 6 = 336$$

ways to award the medals.

5. Consider the different arrangements of the five letters of the word NURSE, such as RUNES and SNUER.

(a) How many arrangements are there?

Solution: Each arrangement is a permutation of the 5 letters, so there are $5! = 120$ arrangements.

(b) How many arrangements begin with a vowel?

Solution: The vowels are U and E, so there are 2 choices for the first letter. The other 4 letters can then be arranged in $4! = 24$ ways. By the Fundamental Principle of Counting, there are $2 \cdot 24 = 48$ such arrangements.

(c) How many arrangements begin with N and end with E?

Solution: With N first and E last, only the middle three letters, U, R, and S, are left to arrange, so there are $3! = 6$ such arrangements.

6. A pizza place offers 10 different toppings. How many different pizzas with exactly 3 different toppings are possible?

Solution: Order does not matter, so each pizza is a 3-combination of the 10 toppings. Therefore, there are

$$_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$

possible pizzas.

7. You are making a playlist of 4 different songs, chosen from your 9 favorites, and the order of the songs matters.

(a) How many playlists are possible?

Solution: Order matters, so each playlist is a 4-permutation of the 9 songs. Therefore, there are $_9P_4 = 9 \cdot 8 \cdot 7 \cdot 6 = 3{,}024$ possible playlists.

(b) Find the answer to part (a) a second way: first choose which 4 songs to use, and then put them in order.

Solution: There are $_9C_4 = \frac{9 \cdot 8 \cdot 7 \cdot 6}{4 \cdot 3 \cdot 2 \cdot 1} = 126$ ways to choose the 4 songs, and then $4! = 24$ ways to put them in order. By the Fundamental Principle of Counting, there are $126 \cdot 24 = 3{,}024$ playlists, the same answer as in part (a). This is just $_nP_k = {}_nC_k \cdot k!$.

8. A nurse educator will show 3 different training videos to new staff, one after the other, chosen from 6 videos on infection control and 5 videos on patient safety. The first video shown must be on infection control.

(a) How many different lineups of 3 videos are possible?

Solution: There are 6 choices for the first video, which must be on infection control. Any of the remaining 10 videos can be shown second, and any of the remaining 9 third. By the Fundamental Principle of Counting, there are $6 \cdot 10 \cdot 9 = 540$ possible lineups.

(b) How many of these lineups include at least one video on patient safety?

Solution: We use complements. A lineup with no video on patient safety is made up entirely of videos on infection control, and there are $6 \cdot 5 \cdot 4 = 120$ of these. Therefore, $540 - 120 = 420$ lineups include at least one video on patient safety.

9. Four friends each tell you the month they were born. How many lists of four birth months, one for each friend, have at least two friends born in the same month?

Solution: Let U be the set of all lists of four birth months, and let A be the set of lists in which at least two friends were born in the same month. Then A^c is the set of lists in which all four months are different. By the Fundamental Principle of Counting, $|U| = 12^4 = 20{,}736$. Order matters and no month is repeated in A^c, so $|A^c| = {}_{12}P_4 = 12 \cdot 11 \cdot 10 \cdot 9 = 11{,}880$. Therefore,

$$|A| = |U| - |A^c| = 20{,}736 - 11{,}880 = 8{,}856.$$

10. Of the 50 volunteers at a hospital, 28 are certified in CPR, 19 are certified in first aid, and 12 are certified in neither.

(a) How many of the volunteers are certified in both?

Solution: Let C be the set of volunteers certified in CPR and F the set of volunteers certified in first aid. Since 12 of the 50 volunteers are certified in neither, $|C \cup F| = 50 - 12 = 38$. By the Inclusion-Exclusion Principle,

$$38 = |C \cup F| = |C| + |F| - |C \cap F| = 28 + 19 - |C \cap F|,$$

so $|C \cap F| = 47 - 38 = 9$.

(b) How many are certified in exactly one of the two?

Solution: Of the 38 volunteers certified in at least one of the two, 9 are certified in both, so $38 - 9 = 29$ are certified in exactly one.

11. How many arrangements of the letters A, B, C, D, and E have A first or B last?

Solution: Let X be the set of arrangements with A first, and let Y be the set of arrangements with B last. In X, the other 4 letters can be arranged in any order, so $|X| = 4! = 24$. In the same way, $|Y| = 4! = 24$. In $X \cap Y$, A is first, B is last, and the middle three letters can be arranged in $3! = 6$ ways. By the Inclusion-Exclusion Principle,

$$|X \cup Y| = |X| + |Y| - |X \cap Y| = 24 + 24 - 6 = 42.$$

12. A patient-safety committee will have a chair and 4 other members with no distinct roles, all chosen from 8 nurses and 4 pharmacists. No one can fill more than one position, and the chair must be a pharmacist.

(a) How many committees are possible?

Solution: There are 4 ways to choose the chair. The 4 other members are then chosen from the remaining 11 people, and order does not matter, so there are ${}_{11}C_4 = 330$ ways to choose them. By the Fundamental Principle of Counting, there are $4 \cdot 330 = 1{,}320$ possible committees.

(b) How many committees have only nurses among the 4 other members?

Solution: There are 4 ways to choose the chair, and then ${}_8C_4 = 70$ ways to choose the 4 other members from the 8 nurses, for a total of $4 \cdot 70 = 280$ committees.

(c) How many committees have at least one pharmacist among the 4 other members?

Solution: These are the committees from part (a) that are not counted in part (b). Using complements, there are $1{,}320 - 280 = 1{,}040$ such committees.

13. How many 4-digit whole numbers (from $1{,}000$ to $9{,}999$) have no repeated digit and are even?

Solution: The last digit must be even. We split into cases by the last digit, and in each case we choose the last digit first, then the first digit, then the two middle digits.

Case 1: The last digit is 0. The first digit can be any of 1 through 9, so there are 9 choices, then 8 choices for the second digit and 7 for the third, for $9 \cdot 8 \cdot 7 = 504$ numbers.

Case 2: The last digit is 2, 4, 6, or 8, so there are 4 choices. The first digit cannot be 0 or the last digit, so there are 8 choices, then 8 choices for the second digit and 7 for the third, for $4 \cdot 8 \cdot 8 \cdot 7 = 1{,}792$ numbers.

The cases are disjoint, so by the Addition Principle there are $504 + 1{,}792 = 2{,}296$ such numbers.

14. Six patients sit in a row of 6 chairs in a clinic waiting room. Two of them, Mr. Lopez and Ms. Grant, do not want to sit next to each other. In how many ways can the six patients be seated?

Solution: We use complements. Without any restriction, the six patients can be seated in $6! = 720$ ways. Now count the seatings in which Mr. Lopez and Ms. Grant sit next to each other. There are 5 pairs of neighboring chairs for them, 2 ways for the two of them to sit in that pair, and $4! = 24$ ways to seat the other four patients, for $5 \cdot 2 \cdot 24 = 240$ seatings. Therefore, there are

$$720 - 240 = 480$$

seatings in which they do not sit next to each other.

Probability

15. A spinner is divided into three equal sections colored red (R), green (G), and blue (B). The spinner is spun twice, and the colors are recorded in order.

(a) Write the sample space by listing its sample points.

Solution: $S = \{\text{RR}, \text{RG}, \text{RB}, \text{GR}, \text{GG}, \text{GB}, \text{BR}, \text{BG}, \text{BB}\}$. By the Fundamental Principle of Counting, $|S| = 3 \cdot 3 = 9$.

(b) Write the event A, "both spins land on the same color," and the event B, "at least one spin lands on red," by listing their sample points.

Solution: $A = \{\text{RR}, \text{GG}, \text{BB}\}$ and $B = \{\text{RR}, \text{RG}, \text{RB}, \text{GR}, \text{BR}\}$.

(c) The sample points are equally likely. Find $P(A)$ and $P(B)$.

Solution: $P(A) = \dfrac{|A|}{|S|} = \dfrac{3}{9} = \dfrac{1}{3}$ and $P(B) = \dfrac{|B|}{|S|} = \dfrac{5}{9}$.

16. A die is weighted so that it lands on 6 with probability $\frac{1}{2}$, and on each of the numbers 1 through 5 with the same probability c.

(a) Use the fact that $\sum_{x \in S} P(x) = 1$ to find c.

Solution: The sample space is $S = \{1, 2, 3, 4, 5, 6\}$, so

$$1 = \sum_{x \in S} P(x) = P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 5c + \frac{1}{2}.$$

Therefore $5c = \frac{1}{2}$, and $c = \frac{1}{10}$.

(b) Let $E = \{1, 3, 5\}$ be the event "the die lands on an odd number." Write $P(E)$ using sigma notation, then evaluate it.

Solution: $P(E) = \sum_{x \in E} P(x) = P(1) + P(3) + P(5) = \dfrac{1}{10} + \dfrac{1}{10} + \dfrac{1}{10} = \dfrac{3}{10}$.

(c) Let $F = \{5, 6\}$ be the event "the die lands on a number greater than 4." Write $P(F)$ using sigma notation, then evaluate it.

Solution: $P(F) = \sum_{x \in F} P(x) = P(5) + P(6) = \dfrac{1}{10} + \dfrac{1}{2} = \dfrac{6}{10} = \dfrac{3}{5}$.

17. Two fair dice are rolled. What is the probability that the larger of the two numbers the dice land on is 3? (If both dice land on the same number, that number counts as the larger one.)

Solution: The larger number is 3 for the sample points $(1,3)$, $(2,3)$, $(3,3)$, $(3,1)$, and $(3,2)$. The 36 sample points are equally likely, so the probability is $\frac{5}{36}$.

18. A hospital has 10 volunteers, 6 of whom are volunteering for the first time and 4 of whom are returning. Three of the volunteers are chosen at random to appear in a newsletter photo.

(a) What is the probability that all 3 are returning volunteers?

Solution: There are ${}_{10}C_3 = 120$ equally likely groups of 3 volunteers. A group of 3 returning volunteers is chosen from the 4 returning volunteers, so there are ${}_4C_3 = 4$ such groups, and the probability is $\frac{4}{120} = \frac{1}{30}$.

(b) What is the probability that exactly 2 are returning volunteers?

Solution: Choose 2 of the 4 returning volunteers and 1 of the 6 first-time volunteers: ${}_4C_2 \cdot {}_6C_1 = 6 \cdot 6 = 36$ groups. The probability is $\frac{36}{120} = \frac{3}{10}$.

(c) What is the probability that at least one is a returning volunteer?

Solution: We use complements. A group with no returning volunteers is made up of 3 of the 6 first-time volunteers, so there are ${}_6C_3 = 20$ such groups, with probability $\frac{20}{120} = \frac{1}{6}$. Therefore, the probability that at least one is a returning volunteer is $1 - \frac{1}{6} = \frac{5}{6}$.

19. A spinner lands on red with probability $\frac{1}{2}$, on blue with probability $\frac{1}{3}$, and on green with probability $\frac{1}{6}$. It is spun twice, and the spins are independent.

(a) What is the probability that both spins land on the same color?

Solution: The event is $\{\text{RR}, \text{BB}, \text{GG}\}$. The spins are independent, so by the Multiplication Rule,

$$P(\text{RR}) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}, \quad P(\text{BB}) = \left(\frac{1}{3}\right)^2 = \frac{1}{9}, \quad P(\text{GG}) = \left(\frac{1}{6}\right)^2 = \frac{1}{36}.$$

These sample points are mutually exclusive, so by the Addition Rule, the probability is

$$\frac{1}{4} + \frac{1}{9} + \frac{1}{36} = \frac{9}{36} + \frac{4}{36} + \frac{1}{36} = \frac{14}{36} = \frac{7}{18}.$$

(b) What is the probability that the two spins land on different colors?

Solution: This is the complement of the event in part (a), so its probability is $1-\frac{7}{18} = \frac{11}{18}$.

(c) What is the probability that one spin lands on red and the other on blue?

Solution: The event is $\{\text{RB}, \text{BR}\}$. By the Multiplication Rule, $P(\text{RB}) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$ and $P(\text{BR}) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$, so by the Addition Rule the probability is $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$.

20. Let A and B be events with $P(A) = 0.6$ and $P(B) = 0.5$. The probability that neither A nor B occurs is 0.2.

(a) Find $P(A \cup B)$.

Solution: The event "neither A nor B occurs" is $(A \cup B)^c$, so $P(A \cup B) = 1 - 0.2 = 0.8$.

(b) Find $P(A \cap B)$.

Solution: By the Inclusion-Exclusion Principle for Events,

$$0.8 = P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - P(A \cap B),$$

so $P(A \cap B) = 1.1 - 0.8 = 0.3$.

(c) Find $P(A \cap B^c)$, the probability that A occurs but B does not.

Solution: The events $A \cap B$ and $A \cap B^c$ are mutually exclusive, and their union is A. By the Addition Rule, $P(A) = P(A \cap B) + P(A \cap B^c)$, so $P(A \cap B^c) = 0.6 - 0.3 = 0.3$.

(d) Are A and B mutually exclusive? Are they independent? Explain.

Solution: A and B are not mutually exclusive, since $P(A \cap B) = 0.3 \neq 0$. They are independent, since $P(A) \cdot P(B) = 0.6 \cdot 0.5 = 0.3 = P(A \cap B)$.

21. A commuter takes a bus, then a train, then a shuttle to get to work. The bus is on time with probability 0.8, the train with probability 0.9, and the shuttle with probability 0.95, independently of each other.

(a) What is the probability that all three are on time?

Solution: By the Multiplication Rule, $0.8 \cdot 0.9 \cdot 0.95 = 0.684$.

(b) What is the probability that at least one of them is late?

Solution: This is the complement of the event in part (a), so its probability is $1 - 0.684 = 0.316$.

(c) What is the probability that the bus is late, but the train and the shuttle are on time?

Solution: The bus is late with probability $1 - 0.8 = 0.2$. By the Multiplication Rule, the probability is $0.2 \cdot 0.9 \cdot 0.95 = 0.171$.

22. A hospital lab mislabels each blood sample with probability 0.002, independently of the other samples. On a busy day, the lab processes 500 samples. What is the probability that at least one of them is mislabeled? (You may use a calculator.)

Solution: Each sample is labeled correctly with probability $1-0.002 = 0.998$. The samples are independent, so by the Multiplication Rule, the probability that all 500 are labeled correctly is $0.998^{500} \approx 0.368$. The event "at least one sample is mislabeled" is the complement of this event, so its probability is

$$1 - 0.998^{500} \approx 1 - 0.368 = 0.632.$$

Even though each sample is very unlikely to be mislabeled, it is more likely than not that at least one of the 500 is.

23. A drawer holds 6 black socks and 4 white socks. You pull out 2 socks at random, one after the other, without looking. What is the probability that the two socks match?

Solution: Let K_1 and K_2 be the events "the first sock is black" and "the second sock is black," and let W_1 and W_2 be the events "the first sock is white" and "the second sock is white." By the General Multiplication Rule,

$$P(K_1 \cap K_2) = P(K_1) \cdot P(K_2 \mid K_1) = \frac{6}{10} \cdot \frac{5}{9} = \frac{1}{3}$$

and

$$P(W_1 \cap W_2) = P(W_1) \cdot P(W_2 \mid W_1) = \frac{4}{10} \cdot \frac{3}{9} = \frac{2}{15}.$$

These events are mutually exclusive, so by the Addition Rule, the probability that the socks match is

$$\frac{1}{3} + \frac{2}{15} = \frac{5}{15} + \frac{2}{15} = \frac{7}{15} \approx 0.467.$$

24. A store records how 200 of its customers pay and whether they use a coupon.

	Uses a coupon	No coupon	Total
Pays by card	48	92	140
Pays with cash	12	48	60
Total	60	140	200

One of the 200 customers is chosen at random. Let K be the event "the customer pays by card" and Q the event "the customer uses a coupon."

(a) Find $P(Q)$, $P(Q \mid K)$, and $P(Q \mid K^c)$.

Solution: $P(Q) = \dfrac{60}{200} = 0.3$. Knowing K shrinks the group to the first row, so $P(Q \mid K) = \dfrac{48}{140} = \dfrac{12}{35} \approx 0.343$. Knowing K^c shrinks the group to the second row, so $P(Q \mid K^c) = \dfrac{12}{60} = 0.2$.

(b) Find $P(K \mid Q)$, and describe it in words.

Solution: Knowing Q shrinks the group to the first column, so $P(K \mid Q) = \dfrac{48}{60} = 0.8$. That is, 80% of the customers who use a coupon pay by card.

(c) Are K and Q independent? Explain.

Solution: No, since $P(Q \mid K) \approx 0.343$ is not equal to $P(Q) = 0.3$. (Alternatively, $P(K) \cdot P(Q) = 0.7 \cdot 0.3 = 0.21$, which is not equal to $P(K \cap Q) = \frac{48}{200} = 0.24$.)

25. A blood bank gets 60% of its units from Donor Center A and 40% from Donor Center B. Screening rejects 2% of the units from Center A and 5% of the units from Center B.

(a) Make a two-way table for a group of $1{,}000$ units, with the donor center for the rows and whether the unit is rejected for the columns.

Solution: Center A supplies 60% of the $1{,}000$ units, or 600 units, and 2% of these, or 12 units, are rejected. Center B supplies the other 400 units, and 5% of these, or 20 units, are rejected.

	Rejected	Accepted	Total
Center A	12	588	600
Center B	20	380	400
Total	32	968	1,000

(b) What is the probability that a randomly chosen unit is rejected?

Solution: $\dfrac{32}{1{,}000} = 0.032$.

(c) A unit is rejected. What is the probability that it came from Center B?

Solution: Knowing that the unit is rejected shrinks the group to the first column: 32 units, 20 of which came from Center B. So the probability is $\dfrac{20}{32} = 0.625$.

26. A fair die is rolled. Let $A = \{1, 2\}$, $B = \{2, 4, 6\}$, and $C = \{5, 6\}$. For each pair of events, decide whether the events are mutually exclusive and whether they are independent.

(a) A and B

Solution: $A \cap B = \{2\} \neq \emptyset$, so A and B are not mutually exclusive. They are independent, since $P(A \cap B) = \frac{1}{6}$ and $P(A) \cdot P(B) = \frac{2}{6} \cdot \frac{3}{6} = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.

(b) A and C

Solution: $A \cap C = \emptyset$, so A and C are mutually exclusive. They are not independent, since $P(A \cap C) = 0$ but $P(A) \cdot P(C) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$.

(c) B and C

Solution: $B \cap C = \{6\} \neq \emptyset$, so B and C are not mutually exclusive. They are independent, since $P(B \cap C) = \frac{1}{6}$ and $P(B) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$.

27. Two fair dice are rolled. Let A be the event "the first die lands on an odd number" and B the event "the numbers the dice land on add up to at least 9." Find $P(B \mid A)$, and compare it with $P(B)$. Are A and B independent?

Solution: The 36 sample points are equally likely, so $P(B \mid A) = \dfrac{|A \cap B|}{|A|}$. The first die lands on 1, 3, or 5 in $3 \cdot 6 = 18$ sample points, so $|A| = 18$, and

$$A \cap B = \{(3, 6), (5, 4), (5, 5), (5, 6)\}.$$

Therefore,

$$P(B \mid A) = \frac{4}{18} = \frac{2}{9}.$$

Also,

$$B = \{(3, 6), (4, 5), (5, 4), (6, 3), (4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)\},$$

so $P(B) = \frac{10}{36} = \frac{5}{18}$. Since $P(B \mid A) = \frac{4}{18} \neq \frac{5}{18} = P(B)$, the events A and B are not independent.

28. A basketball player shoots two free throws. She makes the first one with probability 0.7. If she makes the first, she makes the second with probability 0.8; if she misses the first, she makes the second with probability 0.5.

(a) What is the probability that she makes both?

Solution: Let M_1 and M_2 be the events "she makes the first free throw" and "she makes the second free throw." By the General Multiplication Rule,

$$P(M_1 \cap M_2) = P(M_1) \cdot P(M_2 \mid M_1) = 0.7 \cdot 0.8 = 0.56.$$

(b) What is the probability that she makes exactly one?

Solution: She either makes the first and misses the second, with probability $0.7 \cdot 0.2 = 0.14$, or misses the first and makes the second, with probability $0.3 \cdot 0.5 = 0.15$. These events are mutually exclusive, so by the Addition Rule the probability is $0.14 + 0.15 = 0.29$.

(c) What is the probability that she misses both?

Solution: $0.3 \cdot (1 - 0.5) = 0.3 \cdot 0.5 = 0.15$. As a check, $0.56 + 0.29 + 0.15 = 1$.

29. Each seed in a packet sprouts with probability 0.85, independently of the others. You plant 6 of the seeds.

(a) What is the probability that all 6 sprout?

Solution: This is a binomial experiment with $n = 6$ and $p = 0.85$. Let E_k be the event "exactly k of the seeds sprout." Then

$$P(E_6) = {}_6C_6 \cdot 0.85^6 \cdot 0.15^0 = 0.85^6 \approx 0.377.$$

(b) What is the probability that exactly 5 sprout?

Solution: $P(E_5) = {}_6C_5 \cdot 0.85^5 \cdot 0.15^1 = 6 \cdot 0.85^5 \cdot 0.15 \approx 0.399$.

(c) What is the probability that at most 4 sprout?

Solution: "At most 4" is the complement of $E_5 \cup E_6$, and E_5 and E_6 are mutually exclusive. So the probability is

$$1 - \big(P(E_5) + P(E_6)\big) \approx 1 - (0.399 + 0.377) = 0.224.$$

30. A clinic's front desk calls patients the day before their appointments. Each patient answers the phone with probability 0.6, independently of the others. The desk calls 5 patients.

(a) Explain why this is a binomial experiment. What are n, a success, and p?

Solution: The number of calls, $n = 5$, is fixed, and each patient either answers or does not. The patients answer independently of each other, and the probability that a patient answers is $p = 0.6$ on every call. A success is a patient who answers.

(b) What is the probability that exactly 3 of the patients answer?

Solution: Let E_k be the event "exactly k of the patients answer." Then $P(E_3) = {}_5C_3 \cdot 0.6^3 \cdot 0.4^2 = 10 \cdot 0.216 \cdot 0.16 = 0.3456$.

(c) What is the probability that at least one of the patients answers?

Solution: The complement is "none of the patients answer," which has probability $0.4^5 = 0.01024$. So the probability is $1 - 0.01024 = 0.98976$.

31. A couple plans to have 4 children. Assume that each child is equally likely to be a boy or a girl, independently of the others.

(a) What is the probability that they have exactly 2 girls?

Solution: This is a binomial experiment with $n = 4$ and $p = \frac{1}{2}$, where a success is a girl. Let E_k be the event "exactly k of the children are girls." Then

$$P(E_2) = {}_4C_2 \cdot \left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^2 = \frac{6}{16} = \frac{3}{8}.$$

(b) What is the probability that they have at least one boy and at least one girl?

Solution: The complement is "all girls or all boys." Each of these has probability $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$, and they are mutually exclusive, so the probability is $1 - \frac{2}{16} = \frac{14}{16} = \frac{7}{8}$.